Running Coach App

Full v1 spec — core loop scoped to match Runna's adaptive plan experience

1. v1 Scope

Goal: match Runna's **core loop** — personalized adaptive training plan generation + progress tracking — as closely as possible. Everything outside that loop is explicitly deferred.

In scope for v1

- Race distances: 5K, 10K, half marathon, marathon
- Personalized adaptive plan generation (VDOT + periodization)
- Strava OAuth login (mandatory) + Strava activity sync + manual logging fallback
- Combined adaptation engine: ACWR cap, missed-session reschedule, VDOT recalculation
- Plan versioning (so progress tracking has something to compare against)
- Progress dashboard (4 core views)
- LangGraph coaching-explanation agent
- Pause/resume
- Post-race finish-time capture & VDOT recalibration

Explicitly out of scope for v1 (Phase 2+)

- Strength, mobility, and Pilates sessions
- Real-time audio coaching during runs
- Native watch sync (Garmin/Apple/COROS) — web-only for v1
- Social/community features
- True-beginner plans (run/walk intervals, no pace-zone basis)
- Manual plan editing beyond pause/resume (B-races, day-swapping, “not feeling 100%”)
- Monetization / payment (no billing logic at all in v1)

2. Market context

App	Price	Model
Runna	$19.99/mo or $119.99/yr	Adaptive plans, strength + mobility, watch sync
Garmin Coach	Free (with Garmin watch)	5K/10K/half plans, no marathon
TrainingPeaks	Free tier; Premium from $19.95/mo	Best with a human coach populating it
Human coach	$100-300/mo	Personalised 1:1 coaching

3. Architecture

Layer	Choice	Notes
Frontend	Next.js	Dashboard, plan view, progress charts
Auth	NextAuth (Auth.js) + Strava OAuth	Mandatory for all users (see Section 6)
Backend	FastAPI	Plan generation, adaptation engine, API routes
Database	PostgreSQL	Users, activities, plan versions, sessions
Activity data	Strava API + manual entry fallback	OAuth token per user; matching logic in Section 6
Coaching agent	LangGraph (ReAct-style)	On-demand + proactive triggers, Section 8
Deploy	Vercel (frontend) + Render (backend)	Matches existing project deployments

4. Onboarding inputs

Field	Notes
Goal race distance + date	5K / 10K / half / marathon
Target finish time	Optional. Defaults to current-fitness VDOT if skipped. Feasibility-checked (Section 5).
Current weekly mileage	Anchors week 1 of the generated plan (Section 5).
Training days/week	User picks specific days, not just a count, plus an explicit preferred long-run day.
Current fitness	Recent race time, or a guided time-trial fallback if none exists.
Strava connection	Mandatory login step. If ≥28 days of history exists, backfilled for real ACWR from day one.
Units	Auto-detected from locale/browser, user-overridable.
Timezone	Auto-detected from browser. Drives all “day boundary” logic (missed sessions, etc.).

Feasibility checks

Two checks run at plan creation: target time vs. timeline, and race date vs. each distance's valid plan-length range (5K/10K: 6-10wk, half: 8-14wk, marathon: 12-20wk). Both are **soft warnings only** — the user can proceed anyway. Neither is a hard block.

5. Plan generation logic

Templates

10K/half/marathon share one periodization template (base → build → peak → taper). 5K uses a **separate** template: shorter cycles (6-10 weeks), more speed/VO2max-focused work, less pure base mileage — a marathon-style template applied to a 5K produces a structurally wrong plan, not just a suboptimal one.

Anchoring & peak mileage

Week 1 starts at the user's **reported current weekly mileage**, not a template default. Peak mileage is derived from target-time-implied VDOT combined with distance-specific peak-mileage bands from sports science. The progression curve scales proportionally from the week-1 anchor toward that peak; the ACWR/10%-cap rule (Section 6) governs how fast it's allowed to climb, not how high it goes.

Day assignment

User-selected training days plus an explicit preferred long-run day — not auto-assigned, since auto-assignment frequently conflicts with real schedules.

Time trials

A structured time trial is built into the template roughly every 3-4 weeks, replacing what would otherwise be a tempo session that week — giving the VDOT recalculation trigger a reliable, intentional data point on a known schedule rather than relying only on noisy day-to-day deviation.

6. Adaptation engine

Three triggers run as **one combined engine**, applied in priority order each time it recalculates:

Priority	Trigger	Rule
1 (highest)	ACWR cap	If acute:chronic load ratio > 1.5, cap next week's mileage regardless of template. Overrides the other two.
2	Missed-session reschedule	Push missed key session forward if room before next hard day; otherwise drop it and protect the long run.
3	VDOT recalculation	Only after key sessions (tempo/long run/time trial), only if deviation exceeds ±3% sustained, blended 70% old / 30% new — never a straight overwrite.

Cold-start handling (ACWR)

ACWR needs ~28 days of history to be meaningful. At onboarding, if the connected Strava account has ≥28 days of activity history, it's backfilled and real ACWR runs from day one. Otherwise (new runner, sparse history, or manual-only), a flat **10% week-over-week mileage cap** substitutes for ACWR until 28 days of in-app history accumulate, then handoff is seamless.

“Missed” timing rule

A session isn't marked missed until **end-of-day on the day after** it was scheduled, in the user's local timezone — giving a one-day buffer for normal schedule slips before the reschedule logic triggers. This lines up with the ±1-day Strava matching tolerance in Section 7.

7. Strava integration

Login & cost

Strava OAuth is the **mandatory** login mechanism for every user — there is no separate email/password path. As of the June 2026 developer program change, API access requires an active Strava subscription ($11.99/mo) on the **developer/app-owner's account only** — individual users authorizing via OAuth do not need their own subscription. This one subscription covers the Standard Tier's self-serve limit of up to 10 total connected athletes (you + 9 others), with no per-athlete cost. "Manual logging" (below) is an alternative way to enter activity data after login, not an alternative way to sign in.

Note: if this ever grew past 10 athletes, Standard Tier supports review-based growth up to 9,999 athletes at no extra fee; Extended Access Tier only matters beyond 10,000.

Activity matching

Runna's own approach is explicit linking (start the specific planned workout, or manually link a free run afterward) — not background guessing. The web-app equivalent:

- Default state: a synced Strava activity is **unlinked** until confirmed.
- If exactly one unmatched planned session falls within the date window and distance is roughly consistent, the app shows it as a one-tap **suggested match** — never auto-applied.
- Ambiguous cases (multiple candidates, or none) fall through to a manual picker, matching Runna's "Link Activity" flow.

Duplicate handling

If a manually-logged entry and a Strava-synced activity share the same date and distance within ~5%, they're merged rather than double-counted. The Strava version is kept as source of truth (it carries GPS/pace data the manual entry likely lacks), and the pair only counts once toward mileage and ACWR.

Token expiry / disconnect

If a token refresh fails or the user revokes access in Strava's own settings, the app falls back to manual-logging mode automatically (no hard error), with a visible "reconnect Strava" prompt.

8. Plan versioning & dashboard

Sessions are never overwritten in place. Every adaptation (ACWR cap, reschedule, VDOT recalc) is stored as a new plan version, so the dashboard can always show what was originally planned alongside what it became — without this, the actual-vs-planned dashboard metric below has nothing to compare against.

- Upcoming 7 days of planned sessions
- VDOT trend line over the current training block
- Weekly mileage: actual vs. planned
- Current ACWR status (shown even when it isn't actively capping anything)

9. Coaching-explanation agent (LangGraph)

Aspect	Decision
Trigger timing	ACWR cap fires proactively (inline note, shown immediately). Missed-session reschedule and VDOT recalculation stay on-demand (“why did this change?” click).
Context window	The trigger event plus the last 4 weeks of session data (planned vs. actual, ACWR/VDOT values over that window) — bounded for cost and relevance, not the full training history.
Output format	A short inline text card (2-4 sentences) attached to the relevant session/dashboard element — a single generated note, not a chatbot-style conversation.

This is the main differentiator vs. Runna's silent black-box adjustments — explainable, natural-language reasoning over real recent training data, while staying deliberately scoped (no open-ended chat) so it doesn't balloon into a separate feature.

10. Plan lifecycle

Pause / resume

The only manual plan-editing control in v1. Race date stays fixed during a pause. On resume, the same min/max feasibility check from onboarding re-runs against the remaining time: if it still fits the distance's valid range, the plan compresses to fit; if not, the same soft-warning feasibility flag surfaces rather than silently building an unsafe compressed plan.

Race day / plan completion

On race day, the app prompts for an actual finish time and uses it to recalibrate the user's VDOT baseline going forward — a real race result is the best fitness data point available, and discarding it would mean any future plan starts from a stale estimate. A “what's next” recommendation flow is not built in v1; the result is simply captured and not thrown away.

Prepared for Joel — running coach web app, full v1 spec following grill-me review, June 2026.